

Introduction to Syntactic Parsing



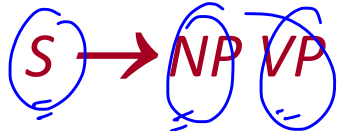
Syntax

Setting out together or arrangement

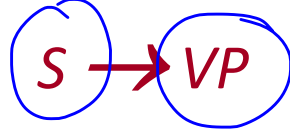
- Refers to the way words are arranged together

Sentence Types

- Declaratives: *A plane left.*



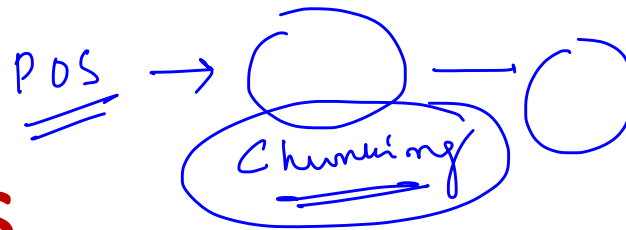
- Imperatives: *Leave!*



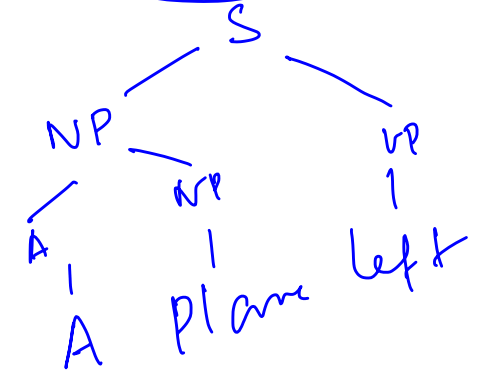
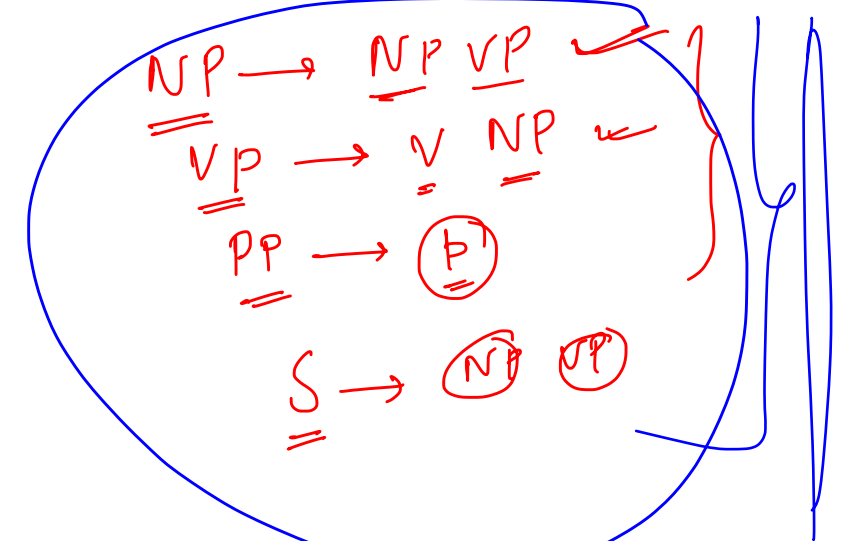
- Yes-No Questions: *Did the plane leave?*



- WH Questions: *When did the plane leave?*



NP VP AP Aux



Two views of linguistic structure:

1. Constituency (phrase structure)

- The basic idea here is that groups of words within utterances can be shown to act as single units

Constituents

- For example, it makes sense to say that the following are all **noun phrases** in English...

Harry the Horse
the Broadway coppers
they

a high-class spot such as Mindy's
the reason he comes into the Hot Box
three parties from Brooklyn

- Why? One piece of evidence is that they can all precede verbs.

Two views of linguistic structure:

1. Constituency (phrase structure)

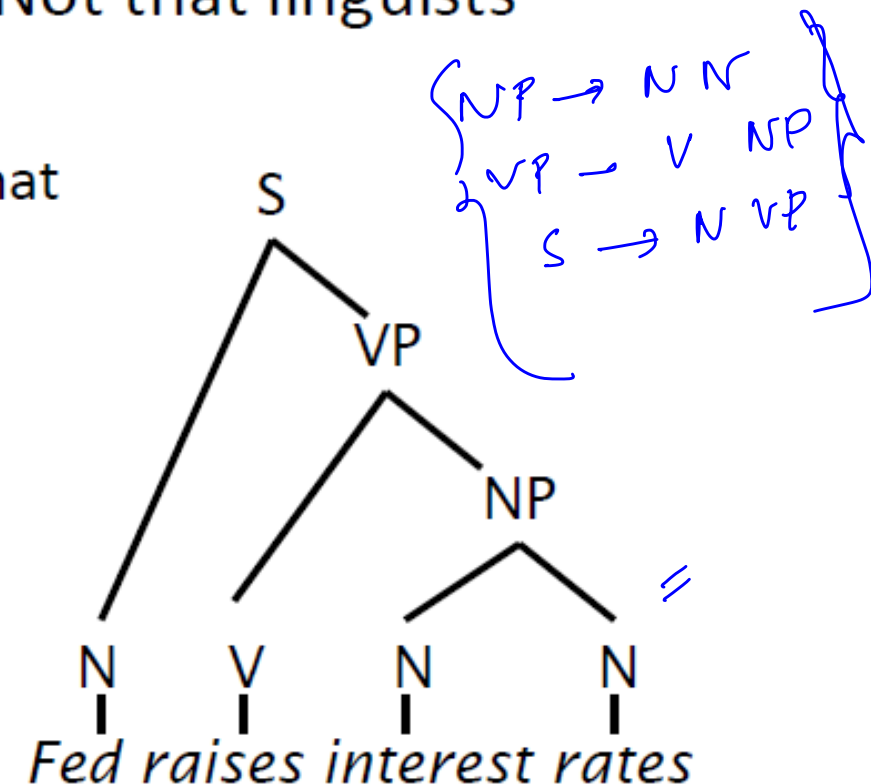
- Phrase structure organizes words into nested constituents.
- How do we know what is a **constituent**? (Not that linguists don't argue about some cases.)

– Distribution: a constituent behaves as a unit that can appear in different places.

- John talked [to the children] [about drugs].
- John talked [about drugs] [to the children].
- *John talked drugs to the children about

– Substitution/expansion/pro-forms:

- I sat [on the box/right of the box/there].



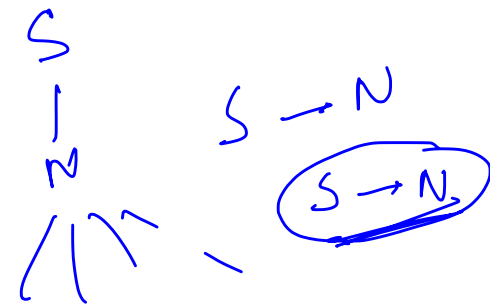
(S (NP Fed) (VP (V raises) (NP (N interest) (N rates))))

S → NP

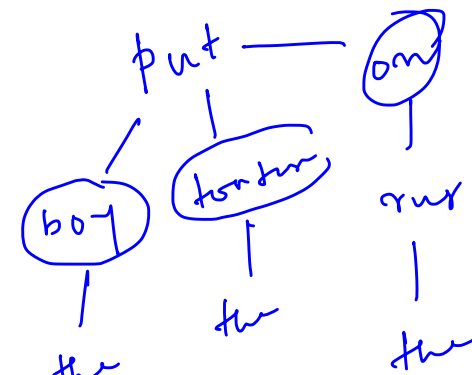
Ramuh

Two views of linguistic structure:

2. Dependency structure

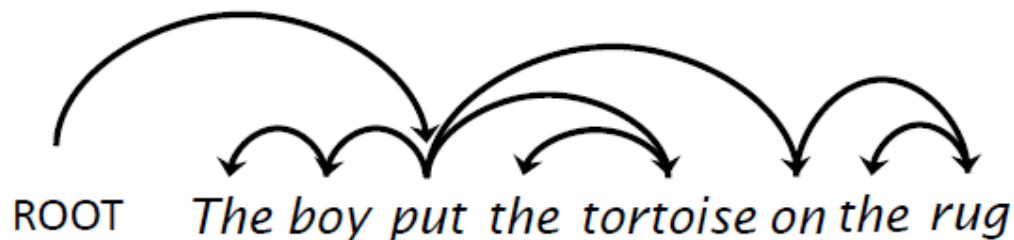


- In CFG-style phrase-structure grammars the main focus is on *constituents*.
- But it turns out you can get a lot done with binary relations among the lexical items (words) in an utterance.
- In a **dependency grammar** framework, a parse is a tree where
 - the nodes stand for the words in an utterance
 - The links between the words represent dependency relations between pairs of words.
 - Relations may be typed (labeled), or not.



dependent head
modifier governor

Sometimes arcs drawn
in opposite direction

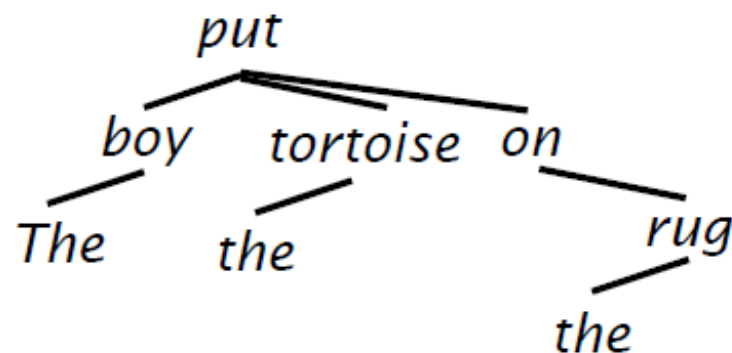
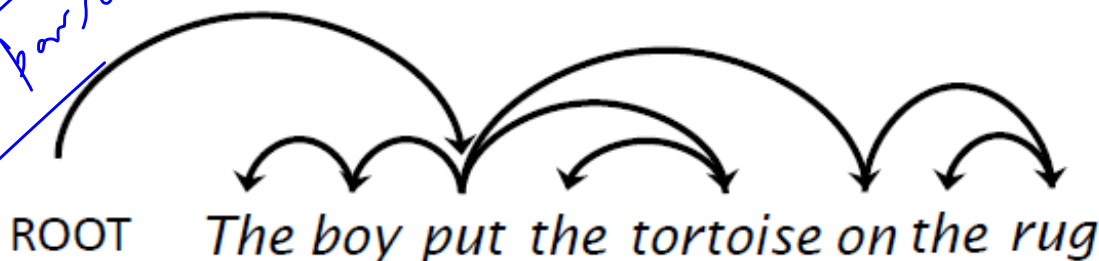


Two views of linguistic structure:

2. Dependency structure

Alternative notations (e.g. rooted tree):

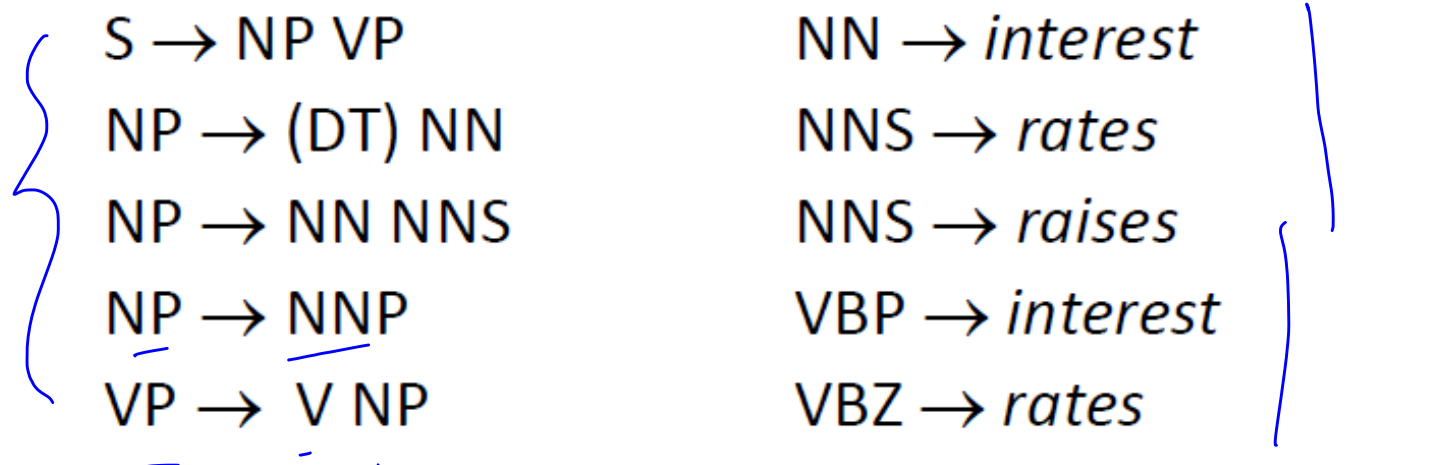
Standard
Dependency parser



The rise of data and statistics:

Pre 1990 (“Classical”) NLP Parsing

- Wrote symbolic grammar (CFG or often richer) and lexicon



$S \rightarrow NP VP$	$NN \rightarrow interest$
$NP \rightarrow (DT) NN$	$NNS \rightarrow rates$
$NP \rightarrow NN NNS$	$NNS \rightarrow raises$
$NP \rightarrow \underline{NNP}$	$VBP \rightarrow interest$
$VP \rightarrow \underline{V} \underline{NP}$	$VBZ \rightarrow rates$

- Used grammar/proof systems to prove parses from words
- This scaled very badly and didn't give coverage.

Fed raises interest rate 0.5% in effort to control inflation.

Minimum grammar: 36 parses

Simple 10 rule grammar: 592 parses

Treebanks [Marcus et al, 1993]

- Treebanks are corpora in which each sentence has been paired with a parse tree (presumably the right one).
- These are generally created
 - By first parsing the collection with an automatic parser
 - And then having human annotators correct each parse as necessary.
- This generally requires detailed annotation guidelines that provide a POS tagset, a grammar and instructions for how to deal with particular grammatical constructions.

The rise of annotated data: The Penn Treebank

[Marcus et al. 1993, *Computational Linguistics*]

((S
(NP-SBJ (DT The) (NN move))
(VP (VBD followed)
(NP
(NP (DT a) (NN round))
(PP (IN of)
(NP
(NP (JJ similar) (NNS increases))
(PP (IN by)
(NP (JJ other) (NNS lenders)))
(PP (IN against)
(NP (NNP Arizona) (JJ real) (NN estate) (NNS loans))))))
(, ,)
(S-ADV
(NP-SBJ (-NONE- *))
(VP (VBG reflecting)
(NP
(NP (DT a) (VBG continuing) (NN decline))
(PP-LOC (IN in)
(NP (DT that) (NN market))))))
(. .)))

PBS

$$\frac{NP-SBJ \rightarrow PT \quad NN}{\left\{ \begin{array}{l} NP \rightarrow N \quad NP = .05 \\ NP \rightarrow NP \quad NP = .7 \\ NP \rightarrow N \quad N \quad NP = .25 \end{array} \right\} \cdot 25} \quad \underline{\underline{1W}}$$

Most well known part is the
Wall Street Journal section of
the Penn TreeBank.
1 M words from the
1987-1989 Wall Street
Journal newspaper.

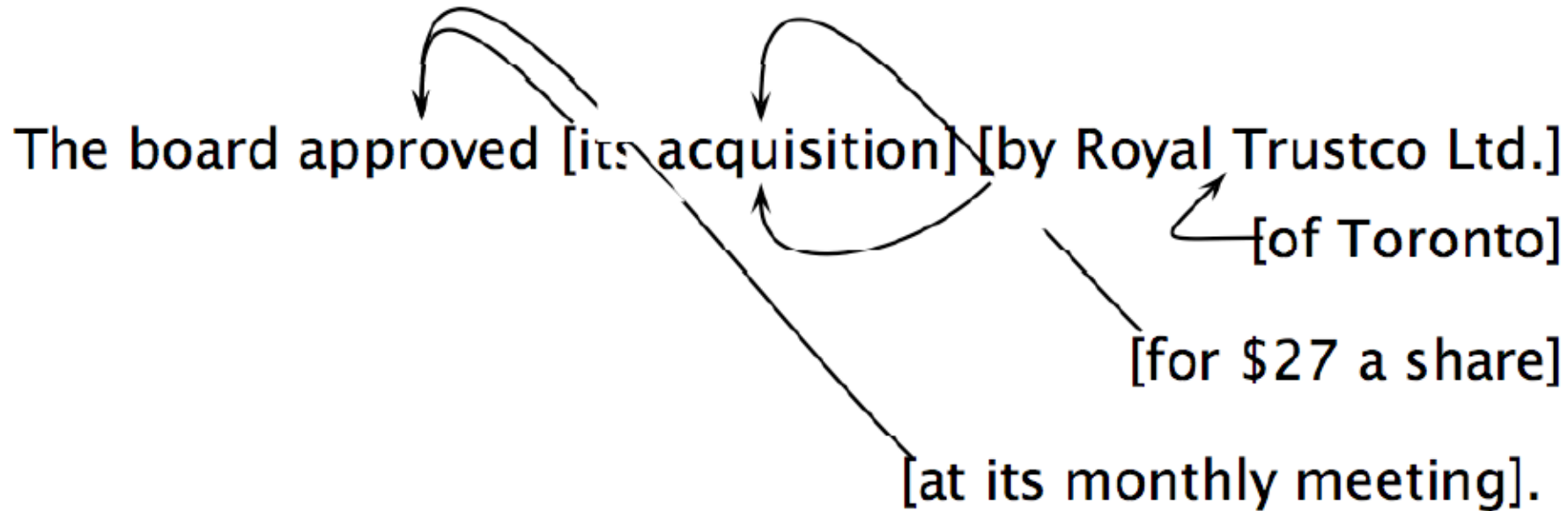
An exponential number of attachments: Attachment ambiguities

- A key parsing decision is how we 'attach' various constituents
PPs, adverbial or participial phrases, infinitives, coordinations, etc

The board approved [its acquisition] [by Royal Trustco Ltd.]
[of Toronto]
[for \$27 a share]
[at its monthly meeting].

An exponential number of attachments: Attachment ambiguities

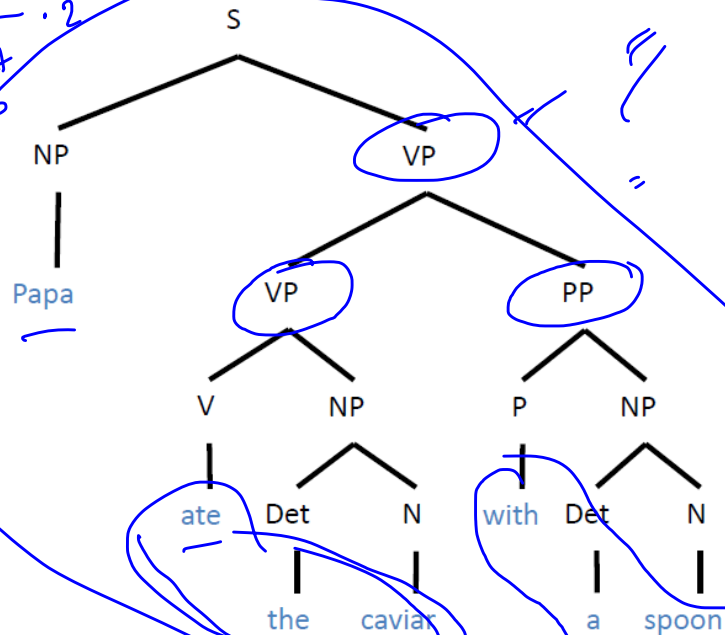
- A key parsing decision is how we 'attach' various constituents
PPs, adverbial or participial phrases, infinitives, coordinations, etc



Two problems to solve:

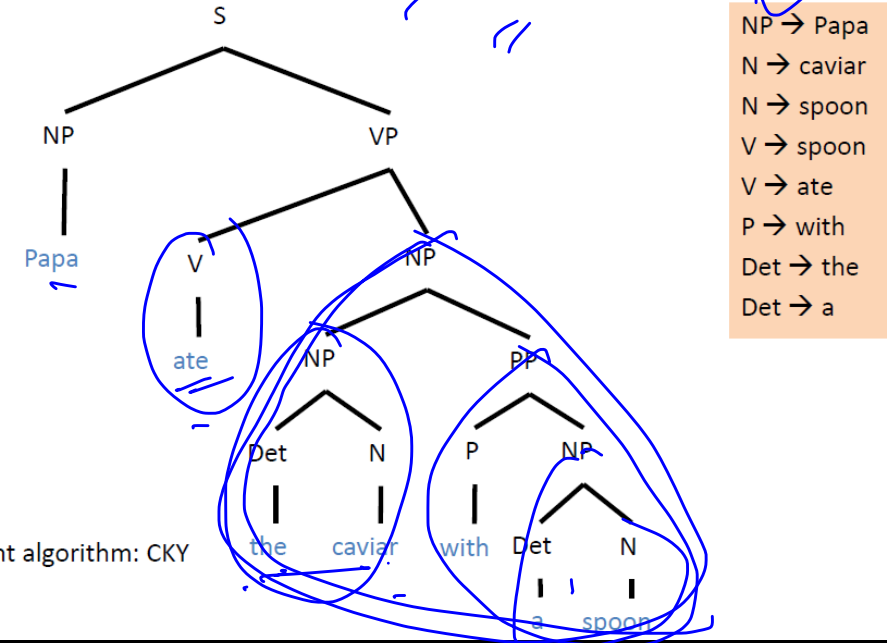
2. Ambiguity - Choosing the correct parse

$S \rightarrow NP VP$
 $NP \rightarrow Det N$
 $NP \rightarrow NP PP$
 $VP \rightarrow V NP$
 $VP \rightarrow VP PP$
 $PP \rightarrow P NP$



$NP \rightarrow Papa$
 $N \rightarrow caviar$
 $N \rightarrow spoon$
 $V \rightarrow spoon$
 $V \rightarrow ate$
 $P \rightarrow with$
 $Det \rightarrow the$
 $Det \rightarrow a$

$S \rightarrow NP VP$
 $NP \rightarrow Det N$
 $NP \rightarrow NP PP$
 $VP \rightarrow V NP$
 $VP \rightarrow VP PP$
 $PP \rightarrow P NP$

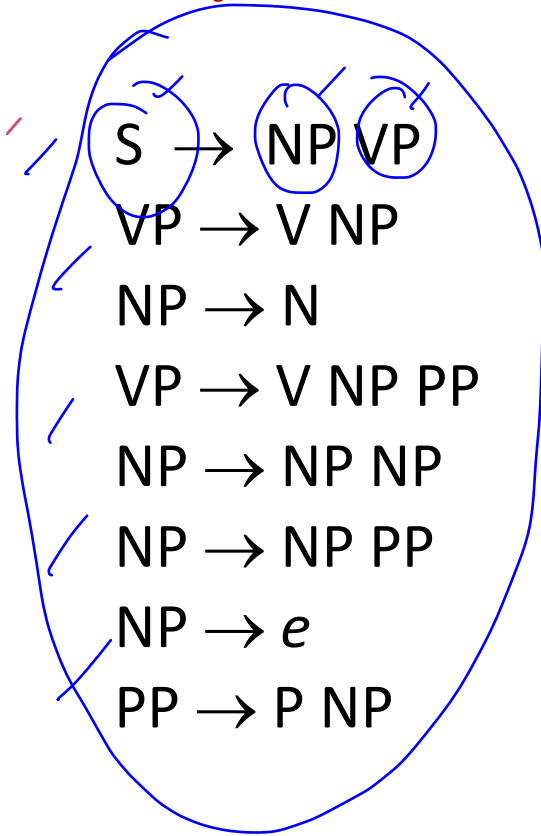


$NP \rightarrow Papa$
 $N \rightarrow caviar$
 $N \rightarrow spoon$
 $V \rightarrow spoon$
 $V \rightarrow ate$
 $P \rightarrow with$
 $Det \rightarrow the$
 $Det \rightarrow a$

→ need an efficient algorithm: CKY

✓ (Probabilistic) Context-Free Grammars

A phrase structure grammar

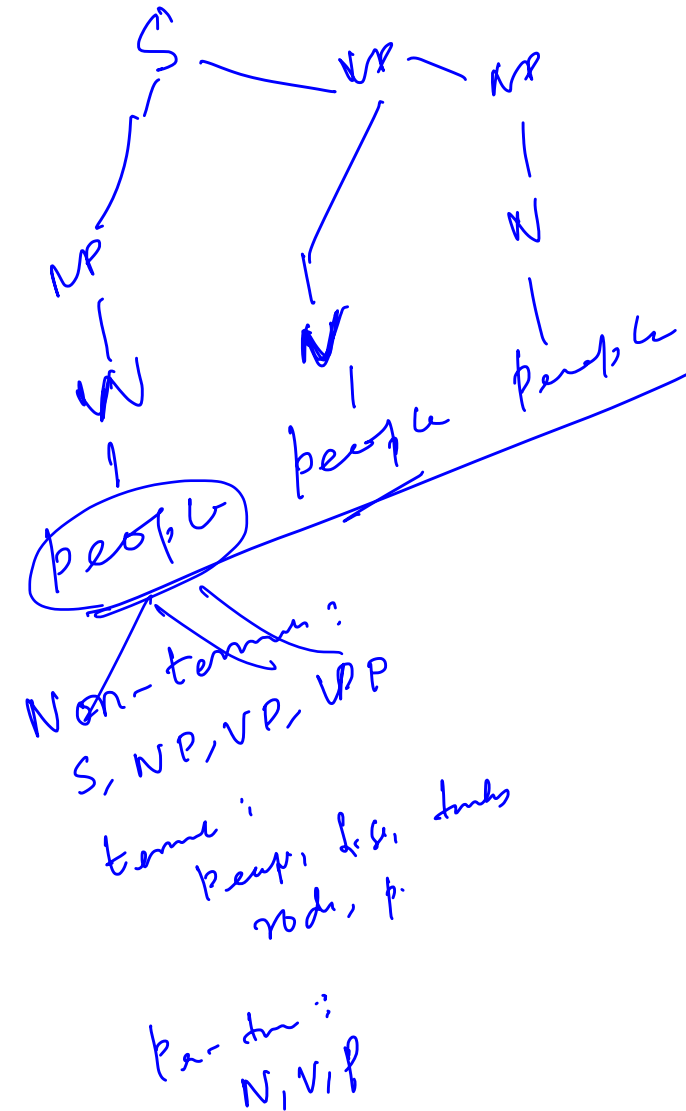
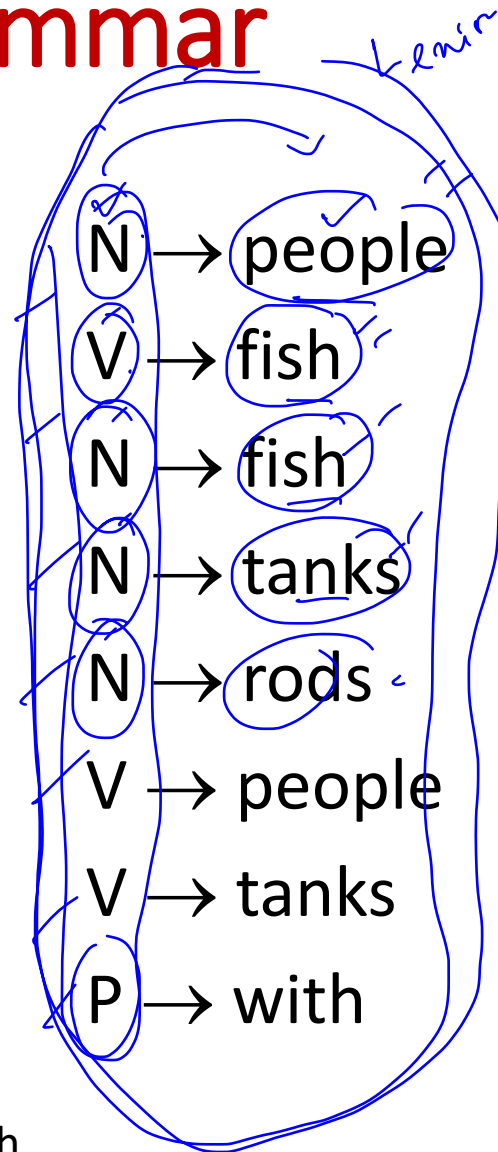


NT
PT
T

people fish tanks

people fish with rods

Ambiguous: People people people, fish fish fish



Phrase structure grammars = context-free grammars (CFGs)

- $G = (T, N, S, R)$

- T is a set of terminal symbols

- N is a set of nonterminal symbols

- S is the start symbol ($S \in N$)

- R is a set of rules/productions of the form $X \rightarrow \gamma$

- $X \in N$ and $\gamma \in (N \cup T)^*$

- A grammar G generates a language L.

NT

P1

Phrase structure grammars in NLP

- $G = (T, C, N, S, L, R)$
 - T is a set of terminal symbols ✓
 - C is a set of preterminal symbols ✓
 - N is a set of nonterminal symbols ✓
 - S is the start symbol ($S \in N$) ✓
 - L is the lexicon, a set of items of the form $X \rightarrow x$
 - $X \in C$ and $x \in T$
 - R is the grammar, a set of items of the form $X \rightarrow \gamma$
 - $X \in N$ and $\gamma \in (N \cup C)^*$
- By usual convention, S is the start symbol, but in statistical NLP, we usually have an extra node at the top (ROOT, TOP)
- We usually write ϵ for an empty sequence, rather than nothing

A phrase structure grammar (empty, unary, binary)

Grammar Rules

$S \rightarrow NP VP$

$VP \rightarrow V NP$

$VP \rightarrow V NP PP$

$NP \rightarrow NP NP$

$NP \rightarrow NP PP$

$NP \rightarrow N$

$NP \rightarrow e$

$PP \rightarrow P NP$

EMPTY fish tanks

people fish EMPTY

Lexicon

$N \rightarrow \textit{people}$

$N \rightarrow \textit{fish}$

$N \rightarrow \textit{tanks}$

$N \rightarrow \textit{rods}$

$V \rightarrow \textit{people}$

$V \rightarrow \textit{fish}$

$V \rightarrow \textit{tanks}$

$P \rightarrow \textit{with}$

Probabilistic/stochastic – context-free grammars (PCFGs)

- $G = (T, N, S, R, P)$
 - T is a set of terminal symbols
 - N is a set of nonterminal symbols
 - S is the start symbol ($S \in N$)
 - R is a set of rules/productions of the form $X \rightarrow \gamma$
 - P is a probability function
 - $P: R \rightarrow [0,1]$

$$\forall X \in N, \sum_{X \rightarrow \gamma \in R} P(X \rightarrow \gamma) = 1$$

10

1 1 1 1

people & time

A PCFG

S → NP VP

1.0 ✓

VP → V NP

0.6 ✓

VP → V NP PP

0.4 ✓

NP → NP NP

0.1 ✓

NP → NP PP

0.2 ✓

NP → N

0.7 ✓

PP → P NP

1.0

N → people

0.5

N → fish

0.2

N → tanks

0.2

N → rods

0.1

V → people

0.1

V → fish

0.6

V → tanks

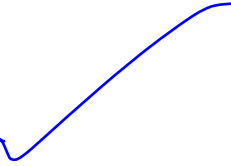
0.3

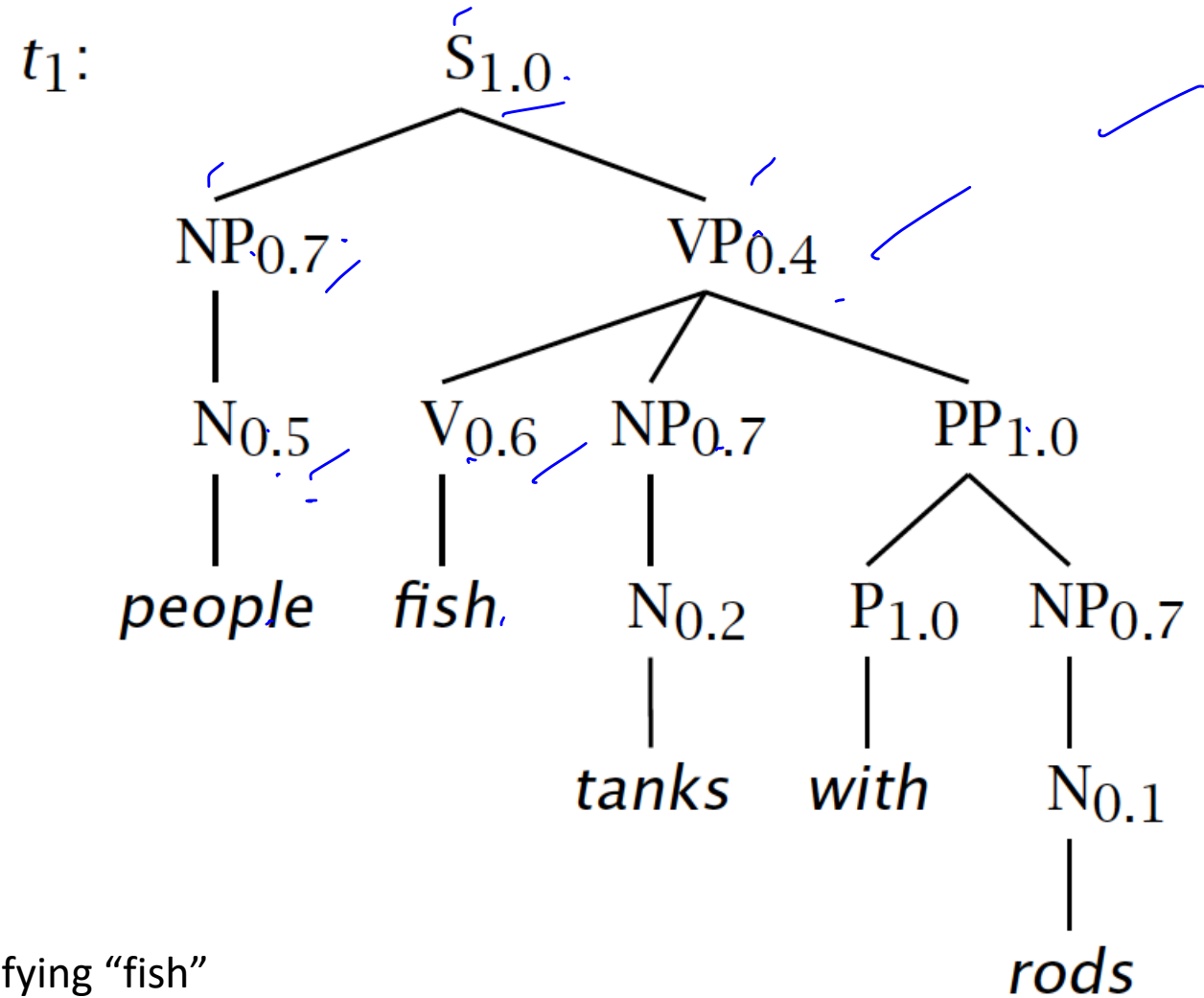
P → with

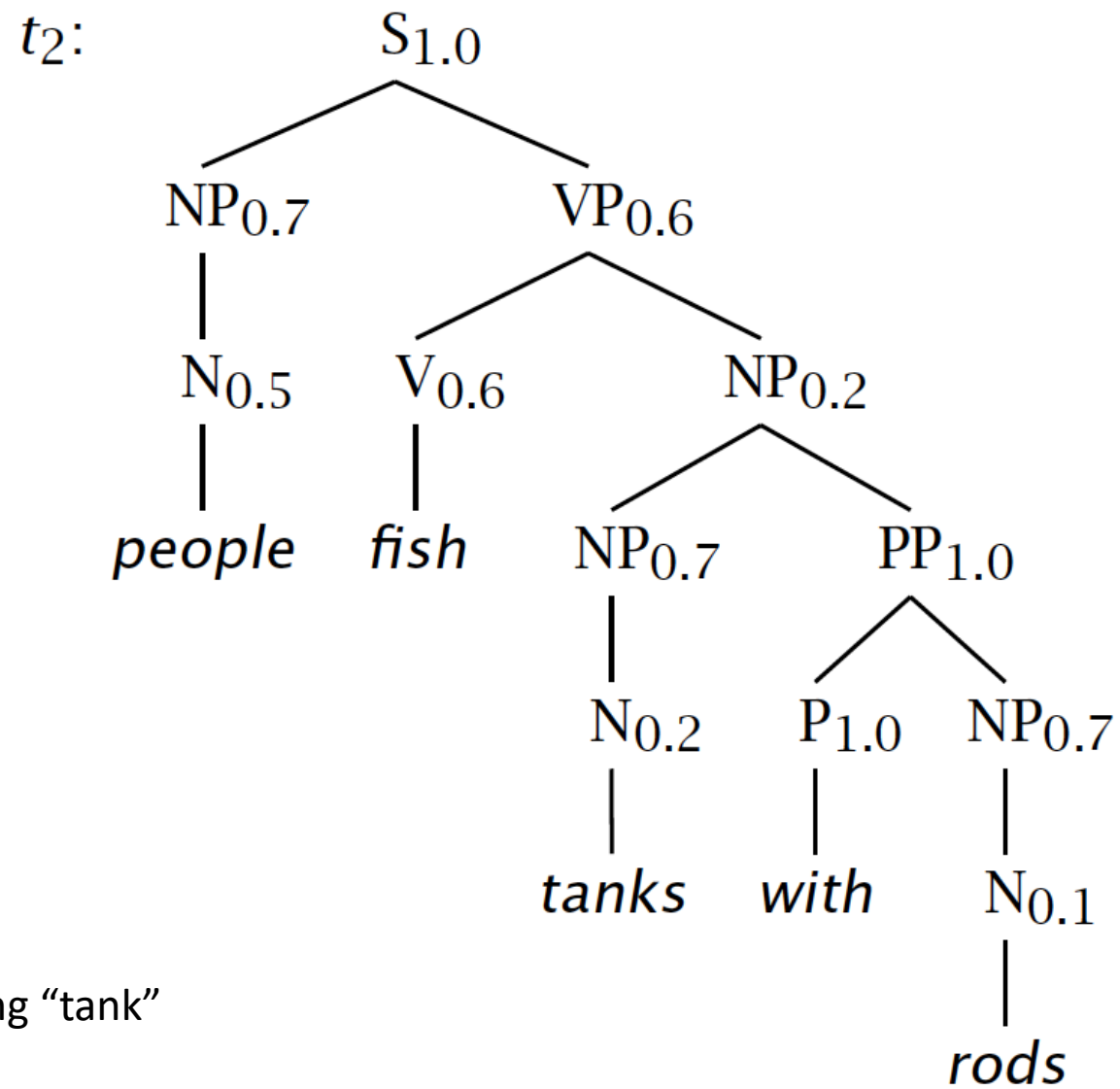
1.0

The probability of trees and strings

- $P(t)$ – The probability of a tree t is the product of the probabilities of the rules used to generate it.
- $P(s)$ – The probability of the string s is the sum of the probabilities of the trees which have that string as their yield

$$P(s) = \sum_t P(s, t) \text{ where } t \text{ is a parse of } s$$
$$= \sum_t P(t)$$






Preposition “with” modifying “tank”

Tree and String Probabilities

- $s = \text{people fish tanks with rods}$

- $P(t_1) = 1.0 \times 0.7 \times 0.4 \times 0.5 \times 0.6 \times 0.7$
 $\times 1.0 \times 0.2 \times 1.0 \times 0.7 \times 0.1$
 $= 0.0008232$

Verb attach

- $P(t_2) = 1.0 \times 0.7 \times 0.6 \times 0.5 \times 0.6 \times 0.2$
 $\times 0.7 \times 1.0 \times 0.2 \times 1.0 \times 0.7 \times 0.1$
 $= 0.00024696$

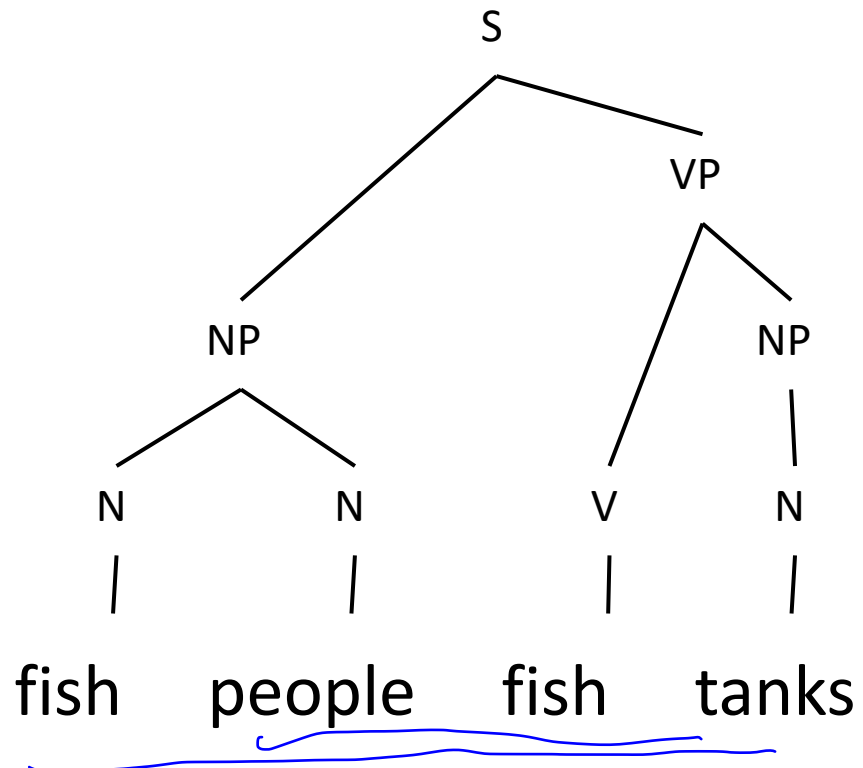
Noun attach

- $P(s) = P(t_1) + P(t_2)$
 $= 0.0008232 + 0.00024696$
 $= 0.00107016$

CKY Parsing

Exact polynomial time parsing of (P)CFGs

Constituency Parsing



PCFG

Rule Prob θ_i

$S \rightarrow \underline{NP VP}$ θ_0 ✓

$NP \rightarrow \underline{NP NP}$ θ_1 ✓

...

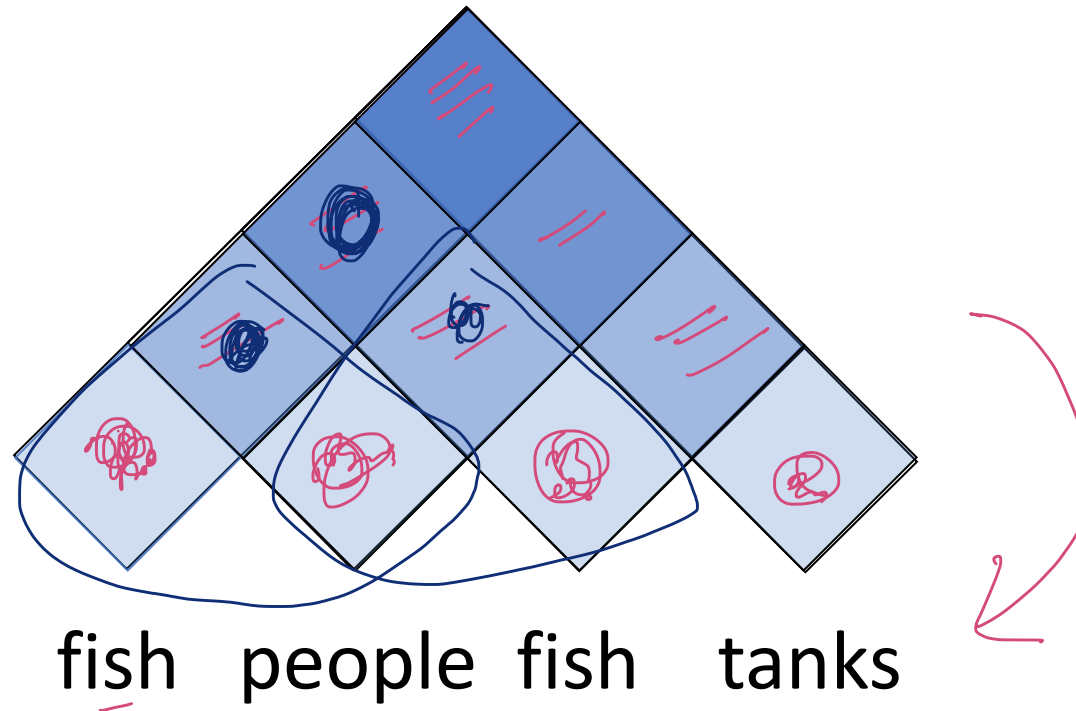
$N \rightarrow \text{fish}$ θ_{42}

$N \rightarrow \text{people}$ θ_{43}

$V \rightarrow \text{fish}$ θ_{44}

...

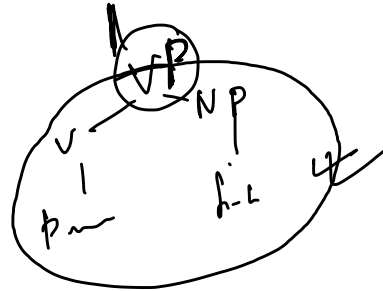
Cocke-Kasami-Younger (CKY) Constituency Parsing



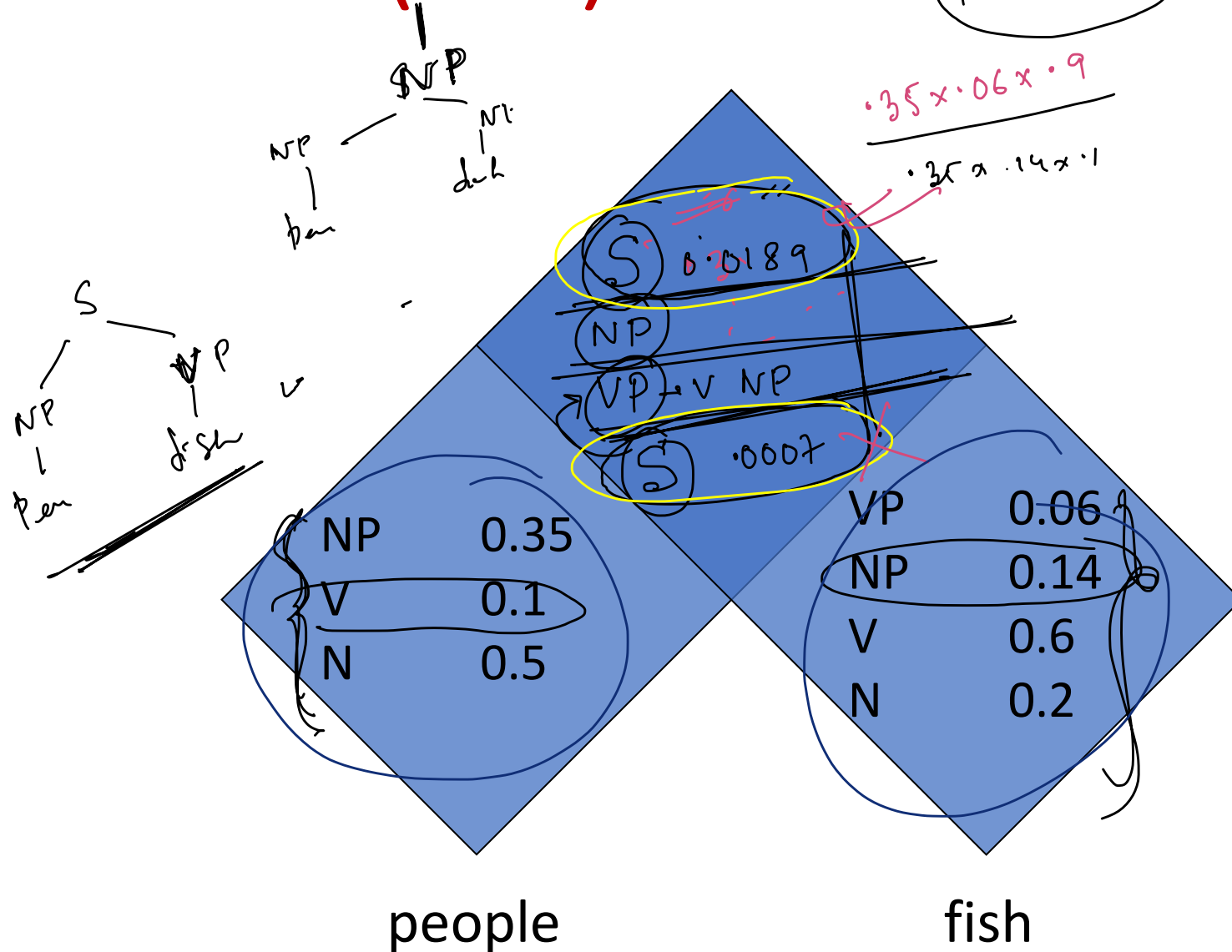
Parse triangle/ chart

Viterbi (Max) Scores

VP → V
@ → VP



$NP \rightarrow NP NP = 0.35 * 0.14 * 0.1 = 0.0049$
 $VP \rightarrow V NP = 0.1 * 0.14 * 0.5 = 0.007$
 $S \rightarrow VP = 0.007 * 0.1 = 0.0007$
 $S \rightarrow NP VP = 0.35 * 0.06 * 0.9 = 0.0189$



S → NP VP	0.9
S → VP	0.1
VP → V NP	0.5
VP → V	0.1
VP → V @VP_V	0.3
VP → V PP	0.1
@VP_V → NP PP	1.0
NP → NP NP	0.1
NP → NP PP	0.2
NP → N	0.7
PP → P NP	1.0

CKY Parsing

A worked example

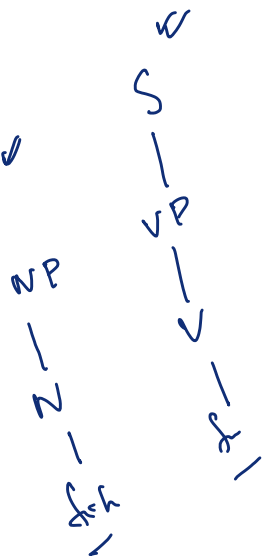
The grammar: Binary, no epsilons,

$S \rightarrow NP VP$	0.9
$S \rightarrow VP$	0.1
$VP \rightarrow V NP$	0.5
$VP \rightarrow V$	0.1
$VP \rightarrow V @VP_V$	0.3
$VP \rightarrow V PP$	0.1
$@VP_V \rightarrow NP PP$	1.0
$NP \rightarrow NP NP$	0.1
$NP \rightarrow NP PP$	0.2
$NP \rightarrow N$	0.7
$PP \rightarrow P NP$	1.0

$N \rightarrow people$	0.5
$N \rightarrow fish$	0.2
$N \rightarrow tanks$	0.2
$N \rightarrow rods$	0.1
$V \rightarrow people$	0.1
$V \rightarrow fish$	0.6
$V \rightarrow tanks$	0.3
$P \rightarrow with$	1.0

	fish	1	people	2	fish	3	tanks	4
0	fish score[0][1]		people score[0][2]		fish score[0][3]		tanks score[0][4]	
1	fish		score[1][2]		score[1][3]		score[1][4]	
2					score[2][3]		score[2][4]	
3							score[3][4]	
4								

$S \rightarrow NP VP$	0.9
$S \rightarrow VP$	0.1
$VP \rightarrow V NP$	0.5
$VP \rightarrow V$	0.1
$VP \rightarrow V @VP_V$	0.3
$VP \rightarrow V PP$	0.1
$@VP_V \rightarrow NP PP$	1.0
$NP \rightarrow NP NP$	0.1
$NP \rightarrow NP PP$	0.2
$NP \rightarrow N$	0.7
$PP \rightarrow P NP$	1.0
$N \rightarrow people$	0.5
✓ $N \rightarrow fish$	0.2
$N \rightarrow tanks$	0.2
$N \rightarrow rods$	0.1
$V \rightarrow people$	0.1
✓ $V \rightarrow fish$	0.6
$V \rightarrow tanks$	0.3
$P \rightarrow with$	1.0



- $S \rightarrow NP VP$ 0.9
- $S \rightarrow VP$ 0.1 ✓
- $VP \rightarrow V NP$ 0.5
- $VP \rightarrow V$ 0.1
- $VP \rightarrow V @VP_V$ 0.3
- $VP \rightarrow V PP$ 0.1
- $@VP_V \rightarrow NP PP$ 1.0
- $NP \rightarrow NP NP$ 0.1
- $NP \rightarrow NP PP$ 0.2
- $NP \rightarrow N$ 0.7 ✓
- $PP \rightarrow P NP$ 1.0
- $N \rightarrow people$ 0.5
- $N \rightarrow fish$ 0.2
- $N \rightarrow tanks$ 0.2
- $N \rightarrow rods$ 0.1
- $V \rightarrow people$ 0.1
- $V \rightarrow fish$ 0.6
- $V \rightarrow tanks$ 0.3
- $P \rightarrow with$ 1.0

	fish	1	people	2	fish	3	tanks	4
0	$N \rightarrow fish\ 0.2$ ✓ $V \rightarrow fish\ 0.6$ ✓ $\bar{NP} \cdot 2 \times .7$ ✓ $\bar{VP} \cdot 1 \times .6$ $\bar{S} \cdot 1 \times .6 \times .1$							
1	fish	$N \rightarrow people\ 0.5$ ✓ $V \rightarrow people\ 0.1$						
2				$N \rightarrow fish\ 0.2$ ✓ $V \rightarrow fish\ 0.6$				
						$N \rightarrow tanks\ 0.2$ ✓ $V \rightarrow tanks\ 0.1$ ✓		

```

// handle unaries
boolean added = true
while added
    added = false
    for A, B in nonterms
        if score[i][i+1][B] > 0 && A->B in grammar
            prob = P(A->B)*score[i][i+1][B]
            if(prob > score[i][i+1][A])
                score[i][i+1][A] = prob
                back[i][i+1][A] = B
                added = true
  
```





$S \rightarrow NP VP$ 0.9
 $S \rightarrow VP$ ✓ 0.1
 $VP \rightarrow V NP$ 0.5
 $VP \rightarrow V$ 0.1
 $VP \rightarrow V @VP_V$ 0.3
 $VP \rightarrow V PP$ 0.1
 $@VP_V \rightarrow NP PP$ 1.0
 $\underline{NP} \rightarrow NP NP$ 0.1
 $NP \rightarrow NP PP$ 0.2
 $NP \rightarrow N$ 0.7
 $PP \rightarrow P NP$ 1.0

 $N \rightarrow people$ 0.5
 $N \rightarrow fish$ 0.2
 $N \rightarrow tanks$ 0.2

 $N \rightarrow rods$ 0.1
 $V \rightarrow people$ 0.1
 $V \rightarrow fish$ 0.6
 $V \rightarrow tanks$ 0.3
 $P \rightarrow with$ 1.0

	0	1	2	3	4
0	$N \rightarrow fish\ 0.2$ $V \rightarrow fish\ 0.6$ $\underline{NP} \rightarrow N\ 0.14$ $\underline{VP} \rightarrow V\ 0.06$ $S \rightarrow VP\ 0.006$	NP VP S $S =$			
1		$N \rightarrow people\ 0.5$ $V \rightarrow people\ 0.1$ $\underline{NP} \rightarrow N\ 0.35$ $VP \rightarrow V\ 0.01$ $S \rightarrow VP\ 0.001$			
2			$N \rightarrow fish\ 0.2$ $V \rightarrow fish\ 0.6$ $\underline{NP} \rightarrow N\ 0.14$ $VP \rightarrow V\ 0.06$ $S \rightarrow VP\ 0.006$		
3				$N \rightarrow tanks\ 0.2$ $V \rightarrow tanks\ 0.1$ $\underline{NP} \rightarrow N\ 0.14$ $VP \rightarrow V\ 0.03$ $S \rightarrow VP\ 0.003$	
4					

$prob = score[begin][split][B] * score[split][end][C] * P(A \rightarrow BC)$
 if ($prob > score[begin][end][A]$)
 $score[begin][end][A] = prob$
 $back[begin][end][A] = new\ Triple(split, B, C)$

$S \rightarrow NP VP$ 0.9
 $S \rightarrow VP$ 0.1
 $VP \rightarrow V NP$ 0.5
 $VP \rightarrow V$ 0.1
 $VP \rightarrow V @VP_V$ 0.3
 $VP \rightarrow V PP$ 0.1
 $@VP_V \rightarrow NP PP$ 1.0
 $NP \rightarrow NP NP$ 0.1
 $NP \rightarrow NP PP$ 0.2
 $NP \rightarrow N$ 0.7
 $PP \rightarrow P NP$ 1.0

 $N \rightarrow people$ 0.5
 $N \rightarrow fish$ 0.2
 $N \rightarrow tanks$ 0.2

 $N \rightarrow rods$ 0.1
 $V \rightarrow people$ 0.1
 $V \rightarrow fish$ 0.6
 $V \rightarrow tanks$ 0.3
 $P \rightarrow with$ 1.0

	fish	1	people	2	fish	3	tanks	4
0								
1	N → fish 0.2 V → fish 0.6 NP → N 0.14 VP → V 0.06 S → VP 0.006	NP → NP NP 0.0049 VP → V NP 0.105 S → NP VP 0.00126 ✓						
2			N → people 0.5 V → people 0.1 NP → N 0.35 VP → V 0.01 S → VP 0.001	NP → NP NP 0.0049 VP → V NP 0.007 S → NP VP 0.0189				
3					N → fish 0.2 V → fish 0.6 NP → N 0.14 VP → V 0.06 S → VP 0.006	NP → NP NP 0.00196 VP → V NP 0.042 S → NP VP 0.00378		
4							N → tanks 0.2 V → tanks 0.1 NP → N 0.14 VP → V 0.03 S → VP 0.003	

```
//handle unaries
boolean added = true
while added
    added = false
    for A, B in nonterms
        prob = P(A->B)*score[begin][end][B];
        if prob > score[begin][end][A]
            score[begin][end][A] = prob
            back[begin][end][A] = B
        added = true
```

```

//handle unaries
boolean added = true
while added
  added = false
  for A, B in nonterms
    prob = P(A->B)*score[begin][end][B];
    if prob > score[begin][end][A]
      score[begin][end][A] = prob
      back[begin][end][A] = B
  added = true

```


$S \rightarrow NP VP$ 0.9
 $S \rightarrow VP$ 0.1
 $VP \rightarrow V NP$ 0.5
 $VP \rightarrow V$ 0.1
 $VP \rightarrow V @VP_V$ 0.3
 $VP \rightarrow V PP$ 0.1
 $@VP_V \rightarrow NP PP$ 1.0
 $NP \rightarrow NP NP$ 0.1
 $NP \rightarrow NP PP$ 0.2
 $NP \rightarrow N$ 0.7
 $PP \rightarrow P NP$ 1.0

 $N \rightarrow people$ 0.5
 $N \rightarrow fish$ 0.2
 $N \rightarrow tanks$ 0.2

 $N \rightarrow rods$ 0.1
 $V \rightarrow people$ 0.1
 $V \rightarrow fish$ 0.6
 $V \rightarrow tanks$ 0.3
 $P \rightarrow with$ 1.0

	fish	1	people	2	fish	3	tanks	4
0	$N \rightarrow fish$ 0.2 $V \rightarrow fish$ 0.6 $NP \rightarrow N$ 0.14 $VP \rightarrow V$ 0.06 $S \rightarrow VP$ 0.006		$NP \rightarrow NP NP$ 0.0049 $VP \rightarrow V NP$ 0.105 $S \rightarrow VP$ 0.0105		$NP \rightarrow NP NP$ 0.0000686 $VP \rightarrow V NP$ 0.00147 $S \rightarrow NP VP$ 0.000882			
1			$N \rightarrow people$ 0.5 $V \rightarrow people$ 0.1 $NP \rightarrow N$ 0.35 $VP \rightarrow V$ 0.01 $S \rightarrow VP$ 0.001		$NP \rightarrow NP NP$ 0.0049 $VP \rightarrow V NP$ 0.007 $S \rightarrow NP VP$ 0.0189			
2					$N \rightarrow fish$ 0.2 $V \rightarrow fish$ 0.6 $NP \rightarrow N$ 0.14 $VP \rightarrow V$ 0.06 $S \rightarrow VP$ 0.006		$NP \rightarrow NP NP$ 0.00196 $VP \rightarrow V NP$ 0.042 $S \rightarrow VP$ 0.0042	
3							$N \rightarrow tanks$ 0.2 $V \rightarrow tanks$ 0.1 $NP \rightarrow N$ 0.14 $VP \rightarrow V$ 0.03 $S \rightarrow VP$ 0.003	
4								

```

for split = begin+1 to end-1
  for A,B,C in nonterms
    prob=score[begin][split][B]*score[split][end][C]*P(A->BC)
    if prob > score[begin][end][A]
      score[begin][end][A] = prob
      back[begin][end][A] = new Triple(split,B,C)
  
```

1, 4
 1, 2 2, 4
 1, 3 3, 4

$S \rightarrow NP VP$ 0.9
 $S \rightarrow VP$ 0.1
 $VP \rightarrow V NP$ 0.5
 $VP \rightarrow V$ 0.1
 $VP \rightarrow V @VP_V$ 0.3
 $VP \rightarrow V PP$ 0.1
 $@VP_V \rightarrow NP PP$ 1.0
 $NP \rightarrow NP NP$ 0.1
 $NP \rightarrow NP PP$ 0.2
 $NP \rightarrow N$ 0.7
 $PP \rightarrow P NP$ 1.0

 $N \rightarrow people$ 0.5
 $N \rightarrow fish$ 0.2
 $N \rightarrow tanks$ 0.2

 $N \rightarrow rods$ 0.1
 $V \rightarrow people$ 0.1
 $V \rightarrow fish$ 0.6
 $V \rightarrow tanks$ 0.3
 $P \rightarrow with$ 1.0

	fish	1	people	2	fish	3	tanks	4
0	$N \rightarrow fish$ 0.2 $V \rightarrow fish$ 0.6 ✓ $NP \rightarrow N$ 0.14 $VP \rightarrow V$ 0.06 $S \rightarrow VP$ 0.006		$NP \rightarrow NP NP$ 0.0049 $VP \rightarrow V NP$ 0.105 $S \rightarrow VP$ 0.0105		$NP \rightarrow NP NP$ 0.0000686 $VP \rightarrow V NP$ 0.00147 $S \rightarrow NP VP$ 0.000882			
1			$N \rightarrow people$ 0.5 $V \rightarrow people$ 0.1 $NP \rightarrow N$ 0.35 $VP \rightarrow V$ 0.01 $S \rightarrow VP$ 0.001		$NP \rightarrow NP NP$ 0.0049 $VP \rightarrow V NP$ 0.007 $S \rightarrow NP VP$ 0.0189		$NP \rightarrow NP NP$ 0.0000686 $VP \rightarrow V NP$ 0.000098 ✓ $S \rightarrow NP VP$ 0.01323	
2					$N \rightarrow fish$ 0.2 $V \rightarrow fish$ 0.6 $NP \rightarrow N$ 0.14 $VP \rightarrow V$ 0.06 $S \rightarrow VP$ 0.006		$NP \rightarrow NP NP$ 0.00196 $VP \rightarrow V NP$ 0.042 $S \rightarrow VP$ 0.0042	
3							$N \rightarrow tanks$ 0.2 $V \rightarrow tanks$ 0.1 $NP \rightarrow N$ 0.14 $VP \rightarrow V$ 0.03 $S \rightarrow VP$ 0.003	
4								

```

for split = begin+1 to end-1
  for A,B,C in nonterms
    prob=score[begin][split][B]*score[split][end][C]*P(A->BC)
    if prob > score[begin][end][A]
      score[begin][end][A] = prob
      back[begin][end][A] = new Triple(split,B,C)
  
```

0 4
 0 1 1 4
 0 2 2 4
 0 3 3 4

$S \rightarrow NP VP$ 0.9
 $S \rightarrow VP$ 0.1
 $VP \rightarrow V NP$ 0.5
 $VP \rightarrow V$ 0.1
 $VP \rightarrow V @VP_V$ 0.3
 $VP \rightarrow V PP$ 0.1
 $@VP_V \rightarrow NP PP$ 1.0
 $NP \rightarrow NP NP$ 0.1
 $NP \rightarrow NP PP$ 0.2
 $NP \rightarrow N$ 0.7
 $PP \rightarrow P NP$ 1.0

 $N \rightarrow people$ 0.5
 $N \rightarrow fish$ 0.2
 $N \rightarrow tanks$ 0.2

 $N \rightarrow rods$ 0.1
 $V \rightarrow people$ 0.1
 $V \rightarrow fish$ 0.6
 $V \rightarrow tanks$ 0.3
 $P \rightarrow with$ 1.0

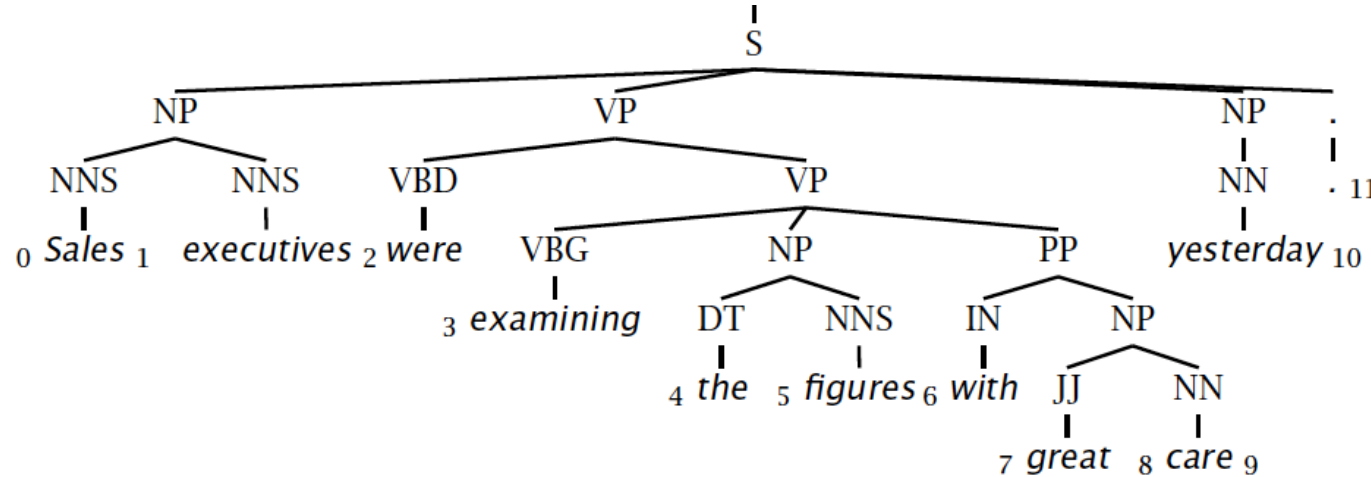
	fish	1	people	2	fish	3	tanks	4
0	$N \rightarrow fish$ 0.2 $V \rightarrow fish$ 0.6 $NP \rightarrow N$ 0.14 $VP \rightarrow V$ 0.06 $S \rightarrow VP$ 0.006		$NP \rightarrow NP NP$ 0.0049 $VP \rightarrow V NP$ 0.105 $S \rightarrow VP$ 0.0105		$NP \rightarrow NP NP$ 0.0000686 $VP \rightarrow V NP$ 0.00147 $S \rightarrow NP VP$ 0.000882		$NP \rightarrow NP NP$ 0.0000009604 $VP \rightarrow V NP$ 0.00002058 $S \rightarrow NP VP$ 0.00018522	
1			$N \rightarrow people$ 0.5 $V \rightarrow people$ 0.1 $NP \rightarrow N$ 0.35 $VP \rightarrow V$ 0.01 $S \rightarrow VP$ 0.001		$NP \rightarrow NP NP$ 0.0049 $VP \rightarrow V NP$ 0.007 $S \rightarrow NP VP$ 0.0189		$NP \rightarrow NP NP$ 0.0000686 $VP \rightarrow V NP$ 0.000098 $S \rightarrow NP VP$ 0.01323	
2					$N \rightarrow fish$ 0.2 $V \rightarrow fish$ 0.6 $NP \rightarrow N$ 0.14 $VP \rightarrow V$ 0.06 $S \rightarrow VP$ 0.006		$NP \rightarrow NP NP$ 0.00196 $VP \rightarrow V NP$ 0.042 $S \rightarrow VP$ 0.0042	
3							$N \rightarrow tanks$ 0.2 $V \rightarrow tanks$ 0.1 $NP \rightarrow N$ 0.14 $VP \rightarrow V$ 0.03 $S \rightarrow VP$ 0.003	
4								

Call buildTree(score, back) to get the best parse

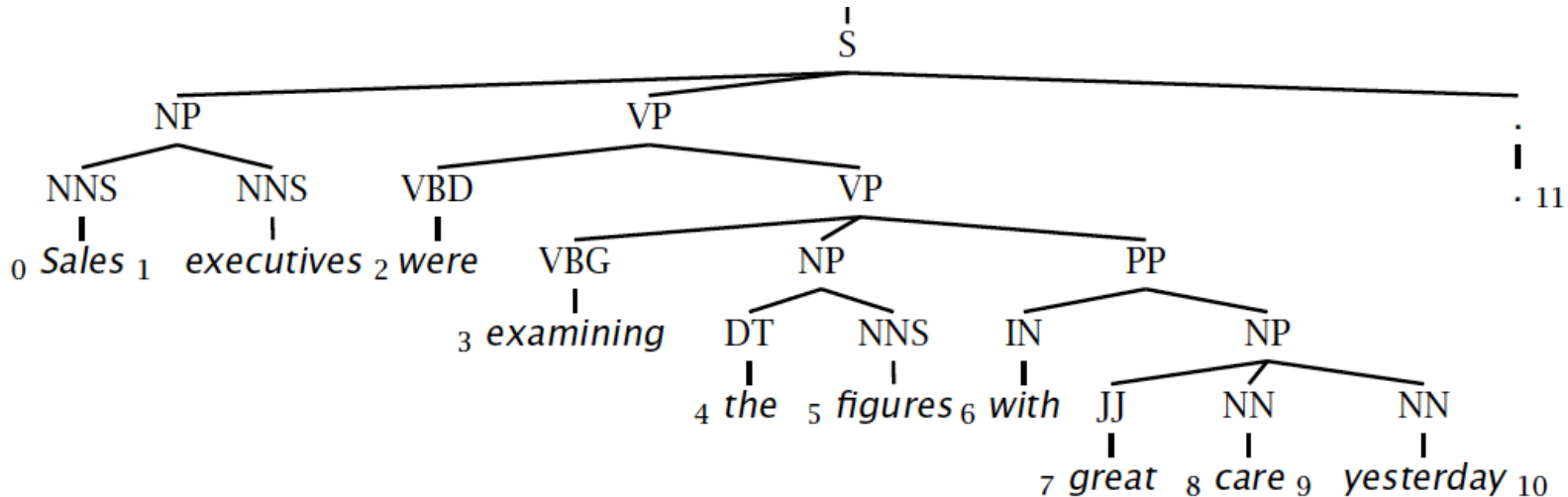
Constituency Parser Evaluation

Evaluating constituency parsing

Gold standard brackets: S-(0:11), NP-(0:2), VP-(2:9), VP-(3:9), NP-(4:6), PP-(6:9), NP-(7,9), NP-(9:10)



Candidate brackets: S-(0:11), NP-(0:2), VP-(2:10), VP-(3:10), NP-(4:6), PP-(6:10), NP-(7,10)



Evaluating constituency parsing

Gold standard brackets:

S-(0:11), NP-(0:2), VP-(2:9), VP-(3:9), NP-(4:6), PP-(6-9), NP-(7,9), NP-(9:10)

Candidate brackets:

S-(0:11), NP-(0:2), VP-(2:10), VP-(3:10), NP-(4:6), PP-(6-10), NP-(7,10)

Labeled Precision	$3/7 = 42.9\%$
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Labeled Recall	$3/8 = 37.5\%$
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LP/LR F1	40.0%
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Tagging Accuracy	$11/11 = 100.0\%$
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